# System identification from input-output data

Technical report · System Lab browser application

**Abstract.** A browser-based workbench simulates first- and second-order processes, imports measured input-output records, estimates an ARX model and evaluates its free-run response on later samples. A documented synthetic second-order record provides a reproducible import example. For this record, ARX(2,2,1) reaches 89.25% validation fit; a higher-order ARX(4,4,1) reaches 98.13% on the same held-out segment.

### 1 Scope and process models

The interface provides a live unit-step response for each process. Gain, time constant, damping ratio and natural frequency can be changed. The two continuous-time transfer functions are:

$$G_1(s) = \frac{K}{\tau s + 1} \tag{1}$$

$$G_2(s) = \frac{K\omega_n^{\,2}}{s^2 + 2\zeta\omega_n s + \omega_n^{\,2}} \tag{2}$$

The first-order response uses an exact zero-order-hold update. The second-order state equations are integrated by fourth-order Runge-Kutta with eight substeps per sample. The noise control adds Gaussian variation to the recorded output, rather than changing the ideal response.

### 2 Imported record

The supplied CSV contains 401 observations from 0.0 to 40.0 s, sampled every 0.1 s. Its columns are time, input and output. The generating process is (2) with $K = 1.8$, $\zeta = 0.45$ and $\omega_n = 2.1$ rad/s. Input levels −1.0, −0.4, 0.4 and 1.0 are held for 0.8–1.5 s; output noise has standard deviation 0.018. The random seed is 1729.

This record was generated by numerical simulation outside the GUI. It is a controlled benchmark for the required CSV-import path, not a physical sensor recording. Changing the GUI's system sliders does not alter imported observations.

### 3 ARX identification

The autoregressive with exogenous input (ARX) model relates the current output to previous outputs and inputs. The orders $n_a$, $n_b$ and $n_k$ set the number of output terms, input terms and initial input delay. For the selected ARX(2,2,1) model:

$$y[k] + a_1 y[k-1] + a_2 y[k-2] = b_1 u[k-1] + b_2 u[k-2] + e[k]$$

The coefficients are found by least squares on the first 70% of the record, with a small diagonal regularizer and pivoted Gaussian elimination. The remaining 30% is predicted recursively. After the split, each prediction uses previous predicted outputs, not measured outputs from the validation segment.

### 4 Estimated ARX(2,2,1) model

The fitted coefficients are $a_1 = -1.736632$, $a_2 = 0.778946$, $b_1 = 0.048327$ and $b_2 = 0.034349$. The resulting discrete transfer function is:

$$G(z) \approx \frac{0.0483 z^{-1} + 0.0343 z^{-2}}{1 - 1.7366 z^{-1} + 0.7789 z^{-2}} \tag{3}$$

# Results and interpretation

Imported second-order dataset · 28.0–40.0 s validation interval

## 5 Validation measure

The residual is r[k] = y[k] − ŷ[k]. RMSE is the root mean square residual in output units. The GUI reports a normalized fit score:

$$\text{Fit} = 100\% \times \left(1 - \frac{\text{RMSE}}{\sigma_{y,\text{val}}}\right) \tag{4}$$

Here $\sigma_{y,\text{val}}$ is the standard deviation of held-out output. Fit measures response agreement, not classification accuracy.

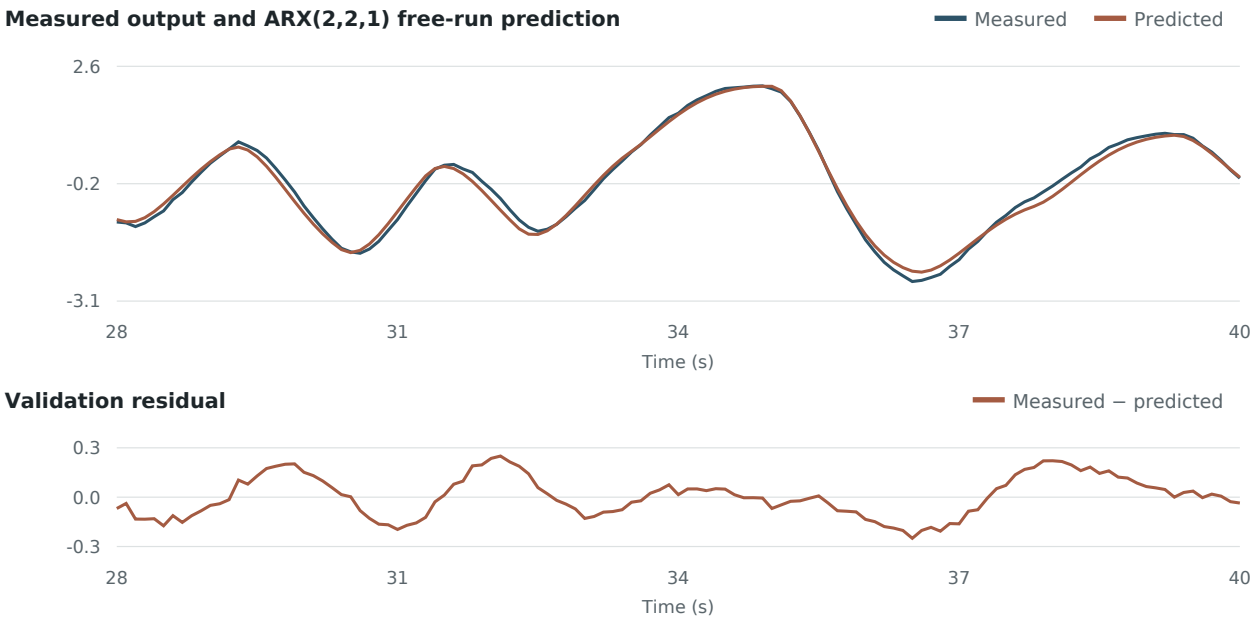



Figure 1. The 121 held-out samples begin at 28.0 s. The residual follows the remaining difference between measured and predicted response.

## 6 Comparison of model orders

| ARX order | Fit (%) | RMSE | Residual lag 1 |
|---|---|---|---|
| (1, 1, 1) | 46.29 | 0.6584 | 0.972 |
| (2, 2, 1) | 89.25 | 0.1318 | 0.951 |
| (4, 4, 1) | 98.13 | 0.0229 | 0.202 |

Table 1. Metrics computed from the rounded CSV using the same chronological split and regularization as the GUI. Residual lag 1 is the correlation between adjacent validation residuals.

ARX(1,1,1) misses substantial dynamics. ARX(2,2,1) captures the main response (89.25% fit; RMSE 0.1318), but its residuals remain strongly correlated (0.951). ARX(4,4,1) reduces error on this held-out interval (98.13% fit; RMSE 0.0229). It does not imply a fourth-order physical system or establish performance under a new input. Noisy lagged outputs can also bias ARX estimates.

**Interactive check.** Increasing τ slows the first-order response; lowering ζ increases second-order overshoot. The user can then change input shape, noise or sampling interval and refit to observe how the validation result changes.